

Same two sections as before, now with the charts that make the equations visible. White noise is the shock. A random walk is those shocks stacked. AR / MA / ARIMA are controlled ways of remembering the past after you decide whether the stack needs to be differenced.

5. White noise and random walk

Every later model is

$$\text{what we observe} = \text{something the model can explain} + \varepsilon_t$$

ε_t is supposed to be **white noise**. If the leftover after a model is not white noise, the model is unfinished. That is why this section comes before ARIMA.

5.1 White noise

A process $\{\varepsilon_t\}$ is white noise when three things hold:

1. Mean does not depend on time (usually zero):

$$\mathbb{E}(\varepsilon_t) = 0$$

1. Variance does not depend on time:

$$\text{Var}(\varepsilon_t) = \sigma^2 < \infty$$

1. Distinct times are uncorrelated:

$$\text{Cov}(\varepsilon_t, \varepsilon_{t-k}) = 0 \quad \text{for all } k \neq 0$$

If the shocks are also Gaussian we write

$$\varepsilon_t \sim \text{i.i.d. } \mathcal{N}(0, \sigma^2)$$

Independence is stronger than uncorrelatedness. For Gaussian variables they coincide,

so in this course “white noise” and “i.i.d. zero-mean shocks” are treated as the same working idea.

The name is from physics: white light has all frequencies equally. White noise has no preferred lag and no preferred frequency. Yesterday’s shock does not help you guess today’s shock.

Characteristics to keep:

- Stationary: constant mean, constant variance, covariance depends only on the lag — and that covariance is zero except at lag 0.
- Unforecastable from its own past. Best forecast of ε_{t+1} is 0.
- ACF: spike at lag 0, then bars inside the $\pm 1.96/\sqrt{T}$ bands.
- PACF: the same.
- The series itself is jitter around a flat line. It does not wander, does not show a season, does not get systematically wilder.

Why it matters. White noise is the standard of leftover ignorance.

- Raw data already white noise → stop. There is no pattern.
- ARIMA residuals are white noise → the univariate structure is finished.
- Residuals still have ACF spikes, a leftover season, or changing variance → the model still has work to do.

Ljung–Box is just a formal way of asking whether that leftover is still correlated with itself.

5.2 Random walk

Start from the same shocks and **add them up**:

$$y_t = y_{t-1} + \varepsilon_t$$

Unwind from a start y_0 :

$$y_t = y_0 + \varepsilon_1 + \varepsilon_2 + \cdots + \varepsilon_t$$

A random walk is the cumulative sum of white noise.

With drift, a constant step is allowed:

$$y_t = \mu + y_{t-1} + \varepsilon_t = y_0 + \mu t + \sum_{i=1}^t \varepsilon_i$$

μt is a straight line under the wander.

The same shocks look completely different once they are stacked:



That figure is the whole distinction. White noise is tied to a mean. A random walk is allowed to take a long excursion and never come back. Add a drift and the excursion has a preferred direction.

Mean.

If there is no drift and $y_0 = 0$, $\mathbb{E}(y_t) = 0$. The average across many imaginary worlds does not wander. One realised path does. Students mix those two things up because the one

path they see looks like a story.

Variance — the failure of stationarity.

$$\text{Var}(y_t) = t \sigma^2$$

The series is allowed to drift farther from its start as time passes. Mean and variance are no longer stable.

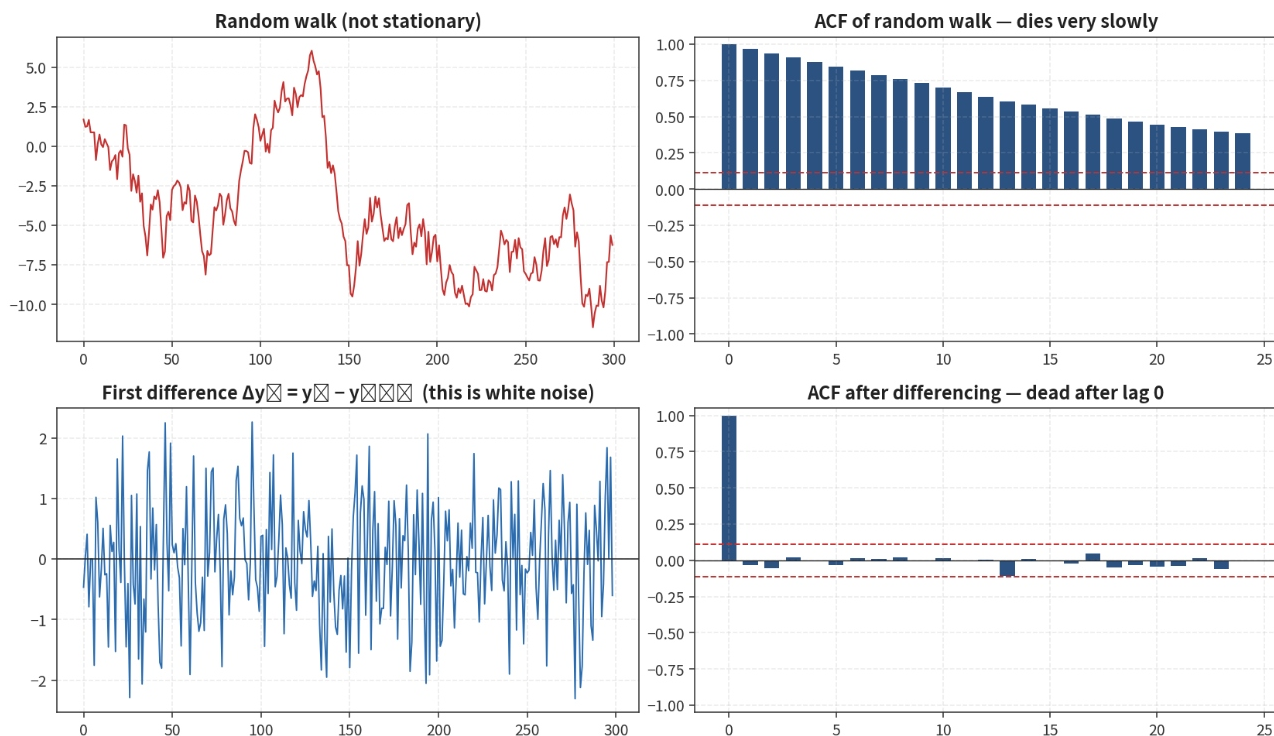
Autocorrelation.

Nearby values share almost the same pile of past shocks, so they are extremely correlated. The sample ACF dies very slowly, often looking like a straight slide down from 1.

First difference — the repair.

$$\Delta y_t = y_t - y_{t-1} = \varepsilon_t$$

The first difference of a driftless random walk is **white noise**. That identity is why differencing is the standard fix, and why a random walk is ARIMA(0, 1, 0).



Top row: wandering series, ACF that will not die.

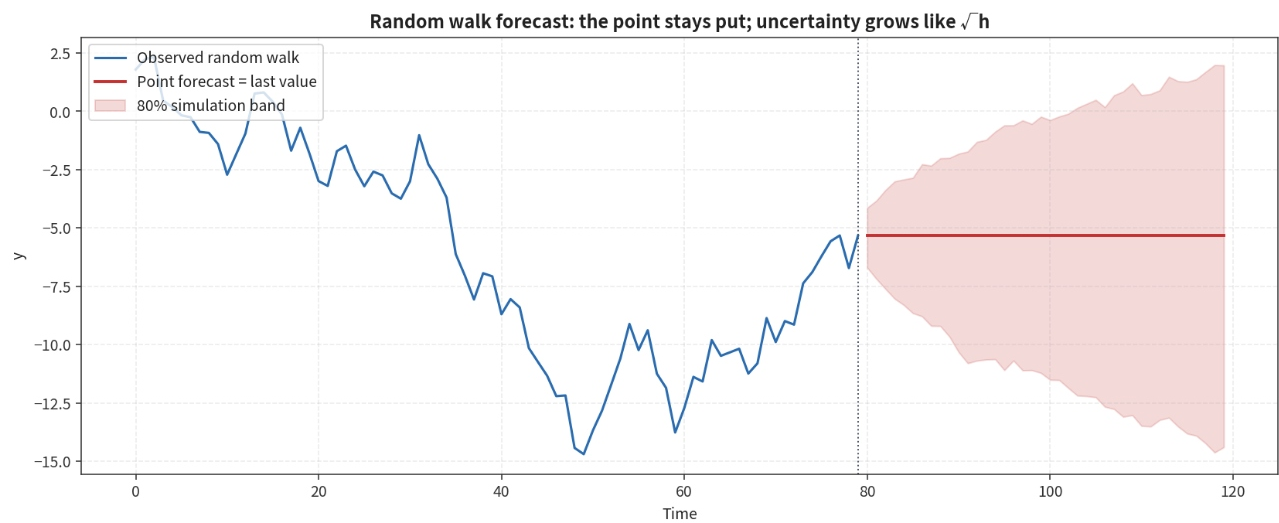
Bottom row: after one difference, jitter around zero and a dead ACF. You did not invent a clever model. You took the extra memory out by subtracting yesterday.

Forecasting a random walk.

The correct *point* forecast is naive:

$$\hat{y}_{t+h} = y_t$$

(or $y_t + \mu h$ with drift). You cannot beat that using only the series' own past. What grows is the **uncertainty**. Variance of the h -step error is $h\sigma^2$, so interval width grows like $\sqrt{h}$.



The red line stays flat. The band opens. That is the honest forecast of a random walk.

Prices of traded assets are the usual metaphor. If the price already contains everything known, the next *change* is a shock. The price itself is close to a random walk; returns are closer to white noise (or white noise with changing variance — ARCH/GARCH, later).

	White noise	Random walk
Equation	$y_t = \varepsilon_t$	$y_t = y_{t-1} + \varepsilon_t$
Stationary?	Yes	No
Mean	Constant	Path wanders; variance grows
ACF	Spike at 0, then ~0	Very slow decay
Best next-step forecast	The mean (usually 0)	The last observed value

First difference	Another noise series	White noise
Look	Jitter around a flat line	Stock-price wandering
Long-horizon forecast	Nothing useful	Point stays put; interval explodes

5.3 How to tell them apart

Never use one number. Stack four views.

1. **Raw plot.** White noise stays in a band. A random walk takes long trips and does not mean-revert.
2. **ACF.** White noise: dead after lag 0. Random walk: slow slide from 1.
3. **Difference once.** If the differenced series looks like white noise, you were looking at a unit-root series of order 1.
4. **ADF and KPSS** (Section 4 in the curriculum). ADF's null is "there is a unit root." KPSS's null is "the series is stationary." A random walk: fail to reject ADF, reject KPSS. White noise: reject ADF's unit-root null, fail to reject KPSS.

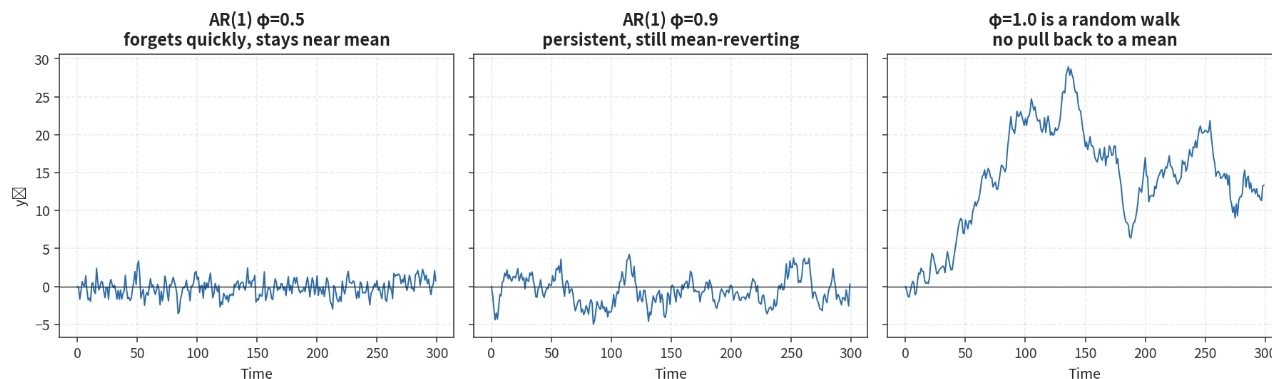
A random walk is the boundary of an AR(1):

$$y_t = \phi y_{t-1} + \varepsilon_t$$

$|\phi| < 1$: pulled back to a mean (stationary AR).

$\phi = 1$: the pullback dies; you have a random walk.

$|\phi| > 1$: explodes.



Left: forgets quickly. Middle: sticky but still tied to zero. Right: no home to return to. That middle-to-right step is the whole “unit root” idea.

6. Time series forecasting models

Construction kit:

- ε_t is always white noise.
 - **AR** remembers past *values* of y .
 - **MA** remembers past *shocks* ε .
 - **ARMA** remembers both, on a stationary series.
 - **ARIMA** differences first, then ARMA.
 - **SARIMA** does the same with a calendar season.
 - **VAR / VMA / VARMA / VARIMA** are the same ideas for several series that influence each other.
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6.1 AR(p)

Today is a linear function of the last p observed values plus a new shock:

$$y_t = c + \phi_1 y_{t-1} + \dots + \phi_p y_{t-p} + \varepsilon_t$$

AR(1):

$$y_t = c + \phi y_{t-1} + \varepsilon_t$$

If $\phi = 0.8$, today keeps 80% of yesterday's deviation from the mean, then adds noise. There is momentum, but as long as $|\phi| < 1$ there is still a long-run mean.

“Auto” means self: ordinary regression on own lags. Errors are not the usual regression errors unless the order is right.

Stationarity. AR(1) needs $|\phi| < 1$. For AR(p) write the polynomial

$$\phi(B) = 1 - \phi_1 B - \dots - \phi_p B^p, \quad By_t = y_{t-1}$$

Every root of $\phi(z) = 0$ must lie **outside** the unit circle.

Fingerprints. PACF cuts off after lag p . ACF tails off (geometric decay, sometimes a dying wave). Once the last p values are in the model, further lags should add nothing — that is what partial autocorrelation measures.

Good for persistent series that slowly forget: slowly moving rates, slowly adjusting demand, many stationary macro series after a transform.

6.2 MA(q)

Today is a linear function of the last q **shocks**, not the last q values:

$$y_t = c + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \dots + \theta_q \varepsilon_{t-q}$$

MA(1):

$$y_t = c + \varepsilon_t + \theta \varepsilon_{t-1}$$

A shock is allowed to spill into the next few periods and then die. Short memory in the *innovations*.

The name is historical. You are not averaging the y 's. You are weighting recent ε 's.

Always stationary (finite combination of white noise). What you need is **invertibility**: shocks recoverable from past y . For MA(1), $|\theta| < 1$. Otherwise two MA models can share an ACF; you want the invertible one.

Fingerprints — mirror of AR. ACF cuts off after lag q . PACF tails off. An MA(q) is uncorrelated with its own past beyond lag q , so ordinary ACF should go quiet after q .

Good when a one-time surprise lasts a few periods: a shipping delay, a two-day backlog after an outage, a measurement error partly corrected next period.

6.3 ARMA(p, q)

Both memories, still on a **stationary** series:

$$y_t = c + \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}$$

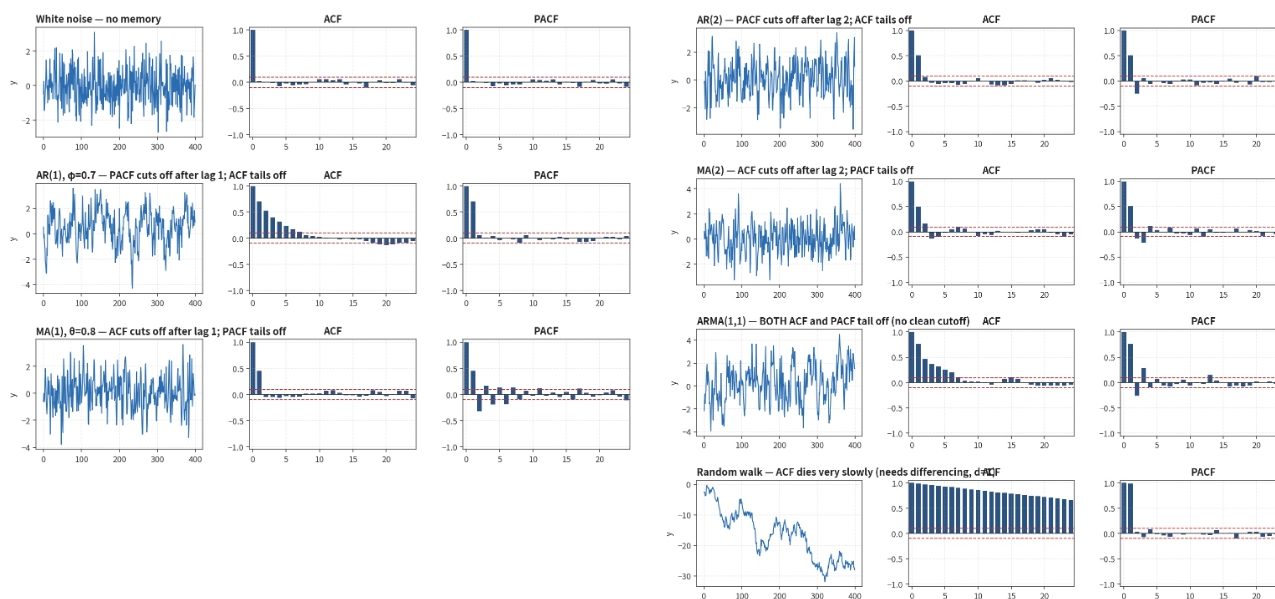
$$\phi(B) y_t = c + \theta(B) \varepsilon_t$$

Use it when neither plot shows a clean cutoff.

Both ACF and PACF tail off. There is no sharp visual rule for both orders. Get a small set of candidates from the plots, fit them, compare AIC/BIC, demand white-noise residuals.

A large ARMA fits the past and fails the future. A small ARMA with white-noise residuals is the goal.

These fingerprints on simulated series:



Read them in order:

- White noise — both plots dead after lag 0. Stop.
- AR(1) — PACF one spike, ACF a decaying tail. $p = 1$.
- MA(1) — ACF one spike, PACF a tail. $q = 1$.
- AR(2) / MA(2) — the cutoff moves to lag 2 on PACF or ACF.

- ARMA(1,1) — both tail off. Do not invent a cutoff.
 - Random walk — ACF refuses to die. Do not fit AR(20). Difference first.
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6.4 ARIMA(p, d, q)

AR and ARMA assume stationarity. Many series are not. The usual failure is random-walk type: the *level* wanders, the *change* is well behaved.

Integration of order d means you difference d times before the series is stationary.

- $d = 0$: already stationary (ARMA).
- $d = 1$: $\Delta y_t = y_t - y_{t-1}$ is stationary. This is the common case.
- $d = 2$: you had to difference the differences (the slope itself wandered).
- $d > 2$: rare; suspect a log, a break, or the wrong model.

ARIMA(p, d, q):

1. difference d times,
2. fit ARMA(p, q) to the differenced series,
3. integrate forecasts back to the original scale.

$$\phi(B) (1 - B)^d y_t = c + \theta(B) \varepsilon_t$$

Special cases:

Process	ARIMA name
White noise	(0, 0, 0)
Random walk	(0, 1, 0)
Random walk with drift	(0, 1, 0) plus a constant
Stationary AR(1)	(1, 0, 0)
Stationary MA(1)	(0, 0, 1)
ARMA(1,1)	(1, 0, 1)

How to choose orders in practice

1. Plot. If the level wanders, suspect $d \geq 1$.
2. Confirm with ADF / KPSS.
3. Difference until the result looks stationary. That is d .
4. On the **differenced** series, read PACF for p and ACF for q .
5. Fit a few neighbours of that (p, d, q) .
6. Choose by AIC/BIC **and** residual white-noise checks.
7. Forecast, then undifference.

A slowly decaying ACF on the *raw* series usually means you have not differenced yet, not that you need a huge p .

6.5 SARIMA — season inside ARIMA

Monthly data remembers last month *and* last year's same month.

SARIMA is two ARIMA models at once: one at lags 1, 2, 3... and one at lags $s, 2s, 3s, \dots$

$$\text{SARIMA}(p, d, q)(P, D, Q)_s$$

- p, d, q : ordinary AR, differences, MA
- P, D, Q : the same three at the seasonal lag
- s : period (12 monthly, 4 quarterly, 7 daily-with-week, 24 hourly-with-day)

Seasonal difference of monthly data:

$$\Delta_{12}y_t = y_t - y_{t-12}$$

You often need **both** $d = 1$ (kill the trend) and $D = 1$ (kill the yearly wave). Classic airline passengers: rise plus growing Julys. After $\Delta\Delta_{12}$, a small ARMA is enough.

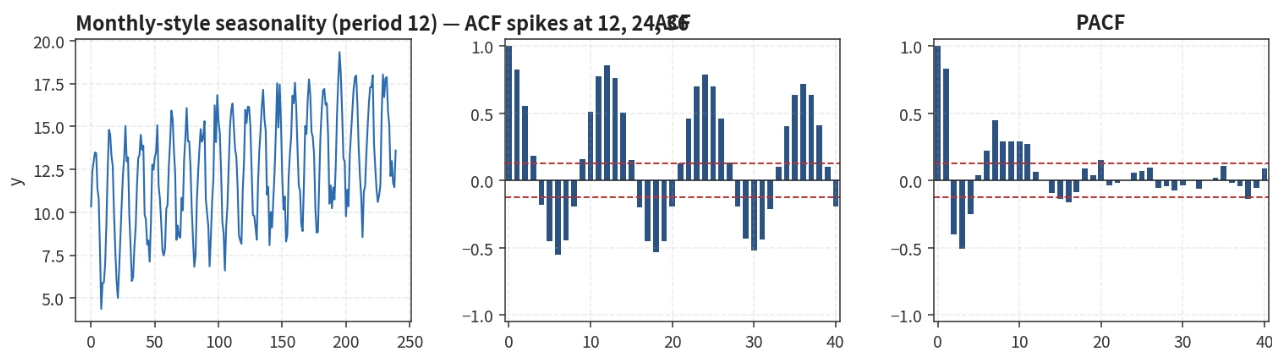
Seasonal ACF/PACF, short version:

- Spike at 12, 24, 36 on ACF that then cuts off $\rightarrow$ seasonal MA.
- Spike at 12 on PACF that then cuts off $\rightarrow$ seasonal AR.

- Slow decay at 12, 24, 36 → you still need seasonal differencing.

SARIMA is still univariate. No temperature, no price. Only the series' own past, including its seasonal past.

Seasonal spikes at 12, 24, 36 look like this:



Those repeating clumps are how you read $s = 12$. They are not a reason to fit AR(36) on raw data.

6.6 The vector family — VAR, VMA, VARMA, VARIMA

Now several series together:

$$\mathbf{y}_t = \begin{pmatrix} y_{1,t} \\ y_{2,t} \\ \vdots \\ y_{k,t} \end{pmatrix}$$

Sales and price and ads. Inflation, the policy rate, unemployment. Each series may depend on its own lags, the other series' lags, and everyone's past shocks.

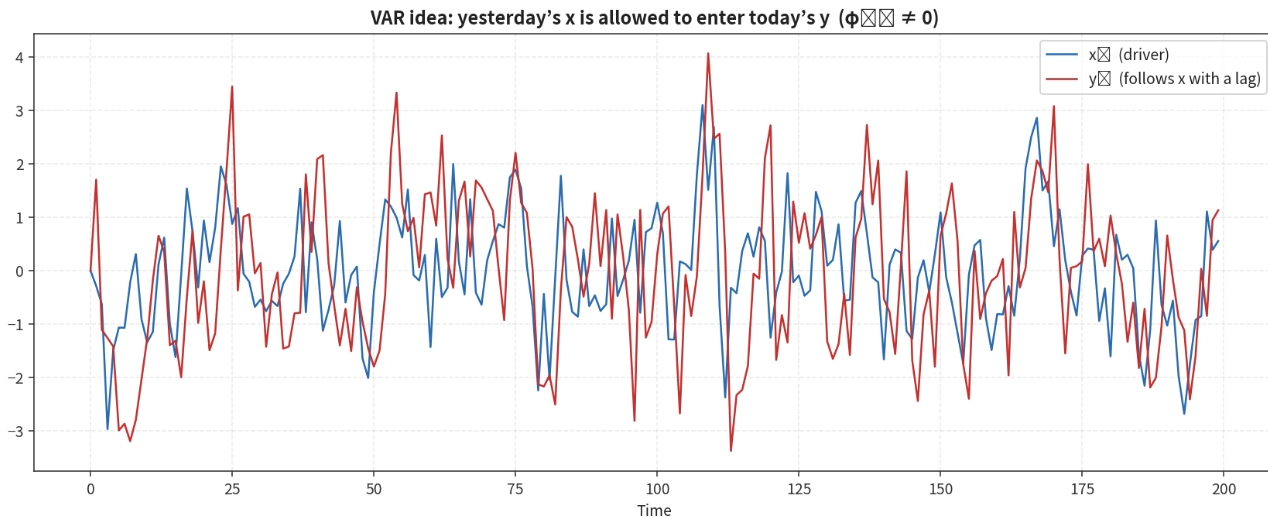
VAR(p)

Each variable is regressed on p lags of all variables. Two series, $p = 1$:

$$\begin{aligned} y_t &= a_1 + \phi_{11}y_{t-1} + \phi_{12}x_{t-1} + \varepsilon_{y,t} \\ x_t &= a_2 + \phi_{21}y_{t-1} + \phi_{22}x_{t-1} + \varepsilon_{x,t} \end{aligned}$$

The off-diagonal coefficients are the new object. ϕ_{12} asks: does yesterday's x help explain today's y ? That is the door to Granger causality, impulse responses, and forecast-error variance decompositions.

A simulated pair where x drives y with a lag:



The red series is not just its own past. It is being pulled around by the blue series. A univariate ARIMA on red would leave that channel on the table.

VAR is the workhorse because each equation can be estimated by OLS. The cost is parameters: k series and lag p means $k^2 p$ slope coefficients. Six monthly series and 12 lags is already hundreds of parameters. Keep the system small, or regularise.

VMA(q)

Each series depends on the last q shocks of all series. Natural language for “an oil shock spills into inflation for two quarters.” In practice VMA is more a representation than a first model. A stationary VAR can be rewritten as an infinite VMA; that rewriting is how impulse responses are computed.

VARMA(p, q)

Vector AR plus vector MA. Flexible, painful to estimate, used less than VAR for the same reason mixed ARMA is used less than people expect.

VARIMA

Difference until the vector process is stationary, then fit a VAR (sometimes a VARMA). Two

habits in practice:

- difference each series from its own ADF/KPSS result, or
- if several non-stationary series wander *together*, do not difference them independently — that throws away the long-run link. That link is **cointegration**, and the model is a VECM. Wholesale and retail prices are the textbook pair.

You do not need VECM to finish this section. You do need to know why “difference everything and run VAR” is not always harmless.

Three questions that place every model

One series or several?

One → AR / MA / ARMA / ARIMA / SARIMA.

Several → VAR / VMA / VARMA / VARIMA.

Stationary?

No → difference (d or D), or de-trend, or log. That is the “I”.

Yes → ARMA / VAR are allowed.

What does it remember?

Past values → AR (VAR).

Past shocks → MA (VMA).

Both → ARMA (VARMA).

Calendar lag s → seasonal block.

Nothing → white noise.

The last value perfectly, $\phi = 1$ → random walk.

Identification sheet (univariate, after stationarity)

What you see	What you try
ACF and PACF dead after lag 0	White noise. Stop.
PACF cuts off after p , ACF tails off	$AR(p)$
ACF cuts off after q , PACF tails off	$MA(q)$
Both tail off	$ARMA(p, q)$, then AIC/BIC
Raw series wanders, ACF dies very slowly	Difference first. Do not read p, q yet.
After one difference, leftover is white noise	Random walk. $ARIMA(0, 1, 0)$
Spikes at 12, 24, ... after ordinary differencing	Seasonal terms or seasonal difference. SARIMA

Then always: residual plot, residual ACF, Ljung–Box. If residuals are not white noise, the order is wrong or a season / break is still in the data.

What not to do yet

- Do not jump to VARIMA because it is on the slide. First see white noise, a random walk, $AR(1)$ and $MA(1)$ on plots.
 - Do not fit ARIMA on a series that still has an obvious seasonal wave. That wave pollutes p and q .
 - Do not treat “auto-ARIMA said (2,1,2)” as understanding. The software ran this checklist. You should be able to look at the ACF/PACF and say whether that pick is plausible.
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Useful next step, same style:

1. **Section 4 — stationarity**, ADF vs KPSS, what each difference does to the ACF.
2. **A full (p, d, q) walk-through** on airline passengers or a series you provide.
3. **Section 2–3 decomposition / STL**, so Holt–Winters and SARIMA have a common picture of trend and season.

Save which one.